

	Exam	ent	Evaluation	Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA3MN207 (P)
Course Title	Statistical inference II
Type of Course	Minor
Semester	III

Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Strong foundation in sampling distributions. Familiarity with simple hypothesis tests.				
Course Summary	This course covers inferential statistics, non-parametric tests, correlation analysis, and regression analysis. Students learn to analyze data using techniques such as ANOVA, Mann-Whitney Test, correlation coefficients, and regression models, enabling them to draw meaningful insights and make informed decisions from statistical data				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental concepts and applications of inferential statistics and critically evaluate ethical implications of statistical methods aligning with human values .	U	C	Instructor-created exams / Quiz
CO2	Evaluate differences in means among multiple independent samples using one-way analysis of variance	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Define and understand the principles of non parametric statistics.	R	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Relate the fundamentals of linear regression and correlation and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Instructor-created exams / Home Assignments
CO5	Explain what regression analysis is and how it differs from correlation analysis.	U	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Apply statistical techniques in software to analyze categorical data effectively	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)				

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)
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COURSE CONTENT

Mo dul e		Content	Hrs (45 +30)	Marks (70)
1		ANOVA	10	20
	1	F Statistic	2	
	2	F Test for Equality of Population Variance	2	
	3	ANOVA	2	
	4	One-Way Analysis of Variance	2	
	5	Two -Way Analysis of Variance	2	
		Sections from References: Unit 1: 19.10,19.10.1,19.10.2[Ref 1] Unit 2:19.10.4[Ref 1] Unit 3: 23.1,23.2 [Ref 1] Unit 4: 23.3[Ref 1] Unit 5: 23.4[Ref 1]		
2		Non Parametric Test	15	15
	5	Introduction to Non Parametric Methods	1	
	6	Advantages and Limitations	1	
	7	Sign Test- one sample	3	
	8	Wilcoxon Signed Rank Test	2	
	9	Mann- Whitney Test	2	
	10	Kruskal- Wallis Test	2	
	11	Single Sample Run Test	2	
	12	Median Test	2	
		Sections from References: Unit 5: 26.2[Ref 2] Unit 6: 26.2.1[Ref 2] Unit 7:26.3 [Ref 2] Unit 8: 26.4[Ref 2] Unit 9: 26.5 [Ref 2] Unit 10: 26.7[Ref 2]		

	Unit 11: 26.8[Ref 2] Unit 12: 26.9[Ref 2]			
3	Correlation Analysis		9	15
	13	Correlation	1	
	14	Types of Correlation	1	
	15	Methods of Studying Correlation	1	
	16	Scatter Diagram Method	2	
	17	Karl Pearson's coefficient of correlation (Concept and Problems)	4	
	Sections from References: Unit 13: 8.1 [Ref 1] Unit 14: 8.1.1[Ref 1] Unit 15: 8.2[Ref 1] Unit 16: 8.3[Ref 1] Unit 17: 8.4[Ref 1]			
4	Regression Analysis		11	20
	18	Introduction to Regression	1	
	19	Linear and Non Linear Regression	1	
	20	Lines of Regression	3	
	21	Coefficients of Regression	3	
	22	Properties of Regression Coefficients	3	
	Sections from References: Unit 18: 9.1 [Ref 1] Unit 19: 9.2[Ref 1] Unit 20: 9.3(9.3.1&9.3.2)[Ref 1] Unit 21: 9.4[Ref 1] Unit 22: 9.4.1,9.5,9.6[Ref 1]			
5	PRACTICUM		30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Scatterplot		
	2	Correlation		
	3	Regression		

	4	Linear correlation coefficient r		
	5	Graphing regression line		
	6	Outliers		
	7	Influential points		
	8	Residual plot		
	Sections from References: Unit 1: 2.4 [Ref 5] Unit 2: 2.4 [Ref 5] Unit 3: 2.4 [Ref 5] Unit 4: 10.1 [Ref 5] Unit 5: 10.2 [Ref 5] Unit 6: 10.2 [Ref 5] Unit 7: 10.2 [Ref 5] Unit 8: 10.2 [Ref 5]			
Books and References: 1. Gupta, S. C.. (2015). Fundamentals of Statistics, Himalaya Publishing House 2. Myra L. Samuels, Jeffrey A. Witmer, Andrew A. Schaffner, Statistics for the Life Sciences ,fifth edition (2016),Pearson Education 3. Prem S. Mann (2016), Introductory Statistics 9 th Edition, Wiley 4. Gupta, S. C. and Kapoor, V. K. (2002). Fundamentals of Mathematical Statistics, 11 th edition, Sulthan Chand, New Delhi 5. Mario F Triola, Elementary Statistics using Excel				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
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CO 1	3	2	-	-	-	2	1	2	-	-	-	3
CO 2	-	-	-	-	-	2	2	2	-	-	-	-
CO 3	-	-	-	-	3	-	2	1	-	-	-	-
CO 4	-	3	-	2	2	1	3	3	2	-	3	-
CO 5	-	-	-	-	-	2	2	2	-	-	-	-
CO 6	-	-	3	-	-	-	1	1	-	3	-	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal	Assignm	Project	End Semester
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	Exam	ent	Evaluation	Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA1MN108 (P)
Course Title	Statistics for critical thinking I
Type of Course	Minor

Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic mathematical knowledge				
Course Summary	This course aims to illustrate the relevance of statistics in social studies by delving into the concept of data, its various forms, generation methods, diverse techniques for summarization and visualization, ultimately fostering a comprehensive understanding.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Identify different types of variables, including categorical, ordinal, interval, and ratio variables, and analyze their characteristics.	U	C	Instructor-created exams / Quiz
CO2	Analyze relationships between variables, including correlations, associations, and causality and analyze data to help entrepreneurial decisions using critical thinking skills..	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Differentiate between qualitative and quantitative data	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Evaluate the advantages and disadvantages of four sampling methods: simple, stratified, cluster, and multistage sampling and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Home Assignments
CO5	Calculate and interpret measures of dispersion, including range, variance, standard deviation, and coefficient of	U	C	One Minute Reflection Writing assignments/